IT AUDITING
IS4101

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Lesson Learning Outcome

By the end of this lesson, you will be able to,

- Understand the purpose of IT governance in organizations
- Explain centralized and distributed IT structures

Content

- IT Governance
- IT Organizational Structure
- Risks & Controls in IT Structure

Information technology governance

- Information Technology (IT) governance is a relatively new subset of corporate governance that focuses on the management and assessment of strategic IT resources
- Key objectives of IT governance –
  - to reduce risk and ensure that investments in IT resources add value to the corporation

IT Governance Controls

- Three IT governance issues
  1. Organizational structure of the IT function
  2. Computer center operations
  3. Disaster recovery planning

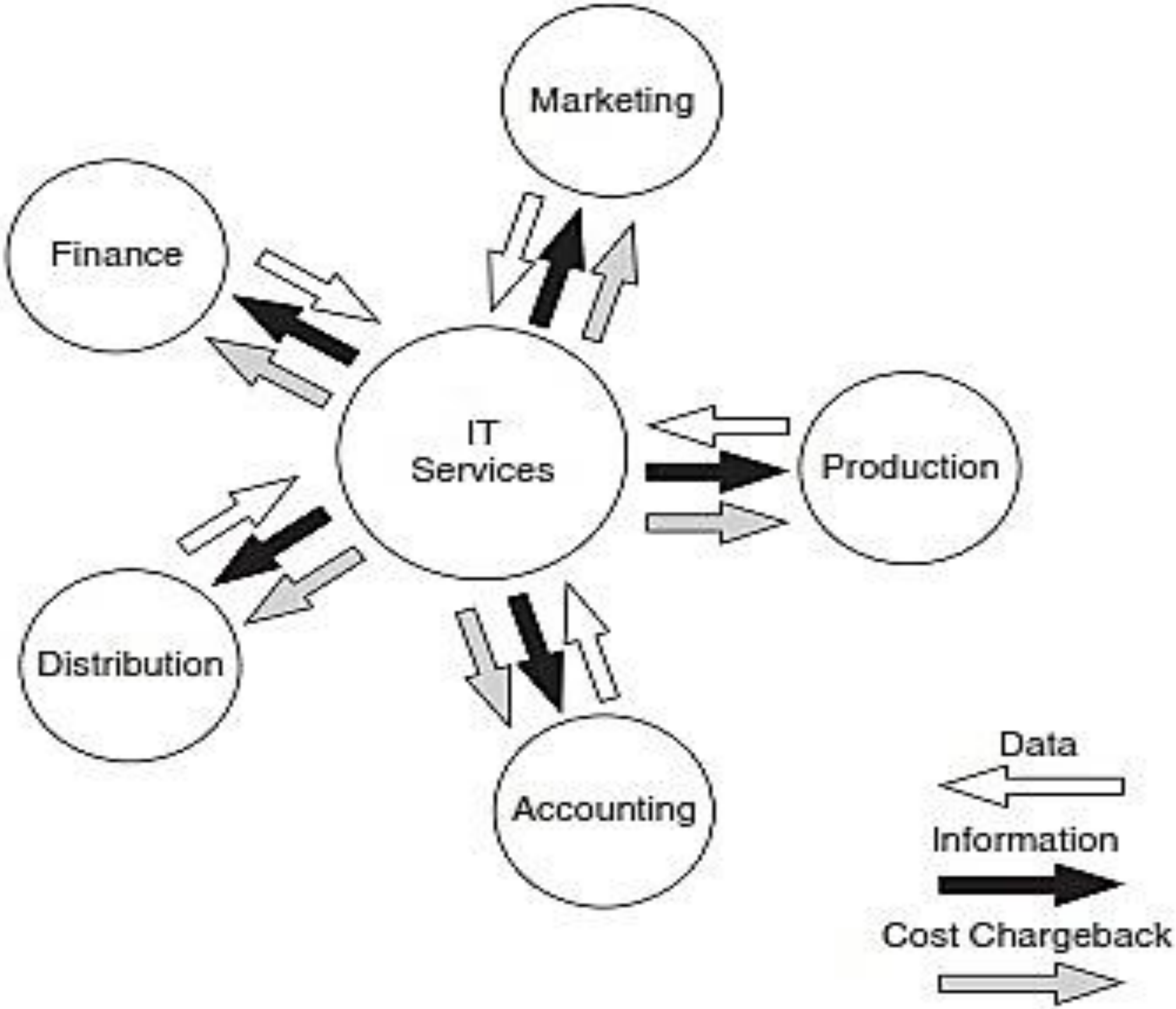
1. Structure of the IT function

1. Structure of the IT function

- The organization of the IT function has implications for the nature and effectiveness of internal controls, which, in turn, has implications for the audit
- Some important control issues related to IT structure can be illustrated through two extreme organizational models
  a) The centralized approach
  b) The distributed approach

a) Centralized Data Processing

- Under the centralized data processing model, all data processing is performed by one or more large computers housed at a central site that serves users throughout the organization
- End users compete for these resources on the basis of need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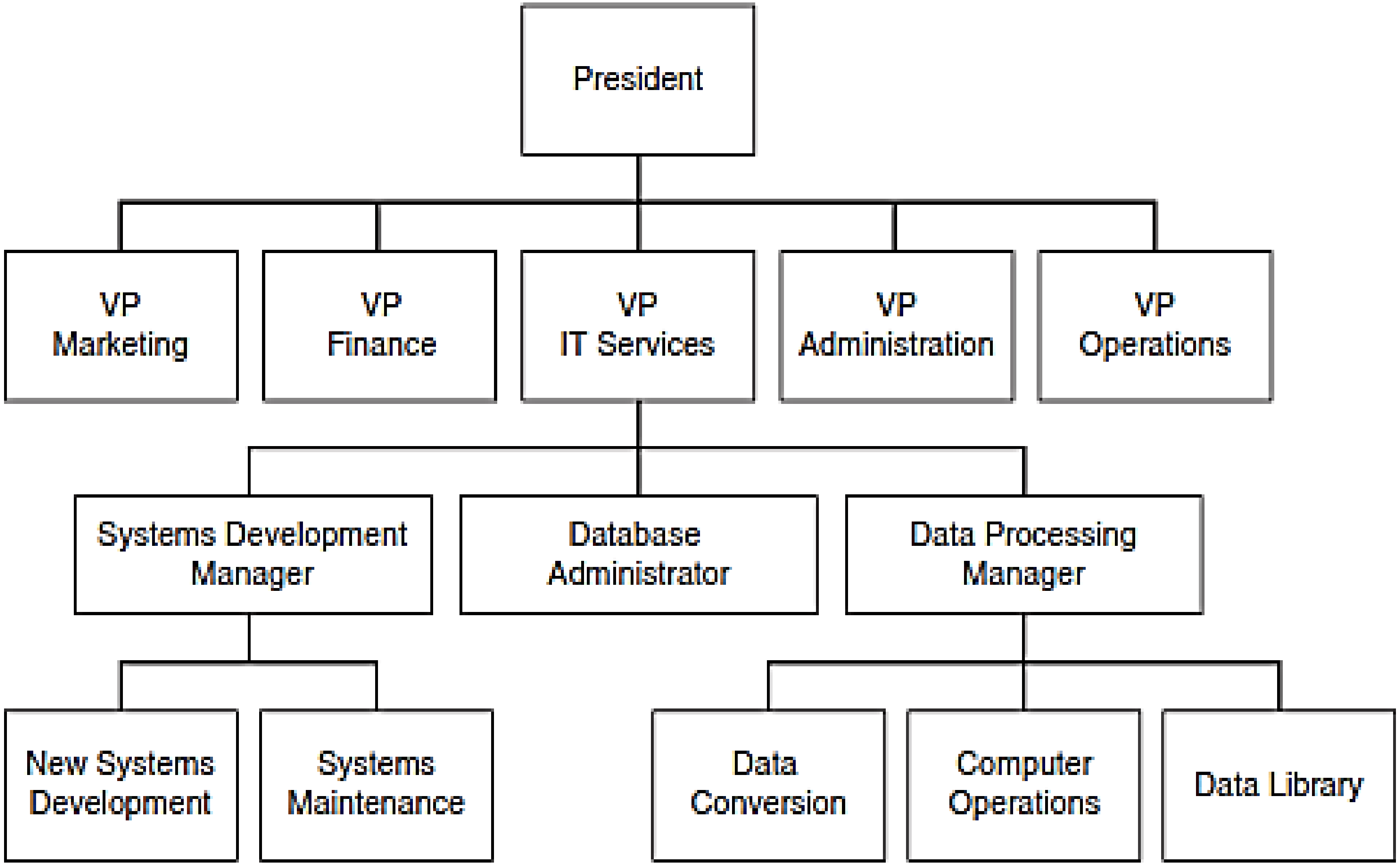


- A centralized IT services structure and shows its primary service areas: database administration, data processing, and systems development and maintenance

a) Centralized Data Processing

- Database administration
- Data processing
  - Data conversion
  - Computer operations
  - Data library
- Systems development and maintenance

Segregation of Incompatible IT Functions

- Operational tasks should be segregated to :
  - Separate transaction authorization from transaction processing
  - Separate record keeping from asset custody
  - Divide transaction processing tasks among individuals such that short of collusion between two or more individuals fraud would not be possible
- The interrelationships among systems development systems maintenance, database administration, and computer operations activities are examined next

Segregation of Incompatible IT Functions

1. **Separating Systems Development from Computer Operations**

- The segregation of systems development (both new systems development and maintenance) and operations activities is of the greatest importance
- The relationship between these groups should be extremely formal, and their responsibilities should not be commingled

Segregation of Incompatible IT Functions

1. **Separating Systems Development from Computer Operations**
   - Systems development and maintenance professionals should create systems for users, and should have no involvement in entering data, or running applications

   (ex.- computer operations)

   - Operations staff should run these systems and have no involvement in their design
   - These functions are inherently incompatible, and consolidating them invites errors and fraud

Segregation of Incompatible IT Functions

2. Separating Database Administration from Other Functions

- Another important organizational control is the segregation of the database administrator (DBA) from other computer center functions
- The DBA function is responsible for a number of critical tasks pertaining to database security, including creating the database schema and user views, assigning database access authority to users, monitoring database usage, and planning for future expansion

Segregation of Incompatible IT Functions

2. Separating Database Administration from Other Functions

- Delegating these responsibilities to others who perform incompatible tasks threatens database integrity

Segregation of Incompatible IT Functions

3. Separating New Systems Development from Maintenance

- Some companies organize their in-house systems development function into two groups: systems analysis and programming
- The systems analysis group works with the users to produce detailed designs of the new systems.
- The programming group codes the programs according to these design specifications

Segregation of Incompatible IT Functions

3. Separating New Systems Development from Maintenance

- Under this approach, the programmer who codes the original programs also maintains the system during the maintenance phase of the systems development life cycle
- Although a common arrangement, this approach is associated with two types of control problems,
    - Inadequate documentation
    - Program fraud

b) Distributed Data Processing

- An alternative to the centralized model is the concept of distributed data processing (DDP)
- The topic of DDP is quite broad, touching upon such related topics as end-user computing, commercial software, networking, and office automation
- DDP involves reorganizing the central IT function into small IT units that are placed under the control of end users
- The IT units may be distributed according to business function, geographic location, or both

b) Distributed Data Processing

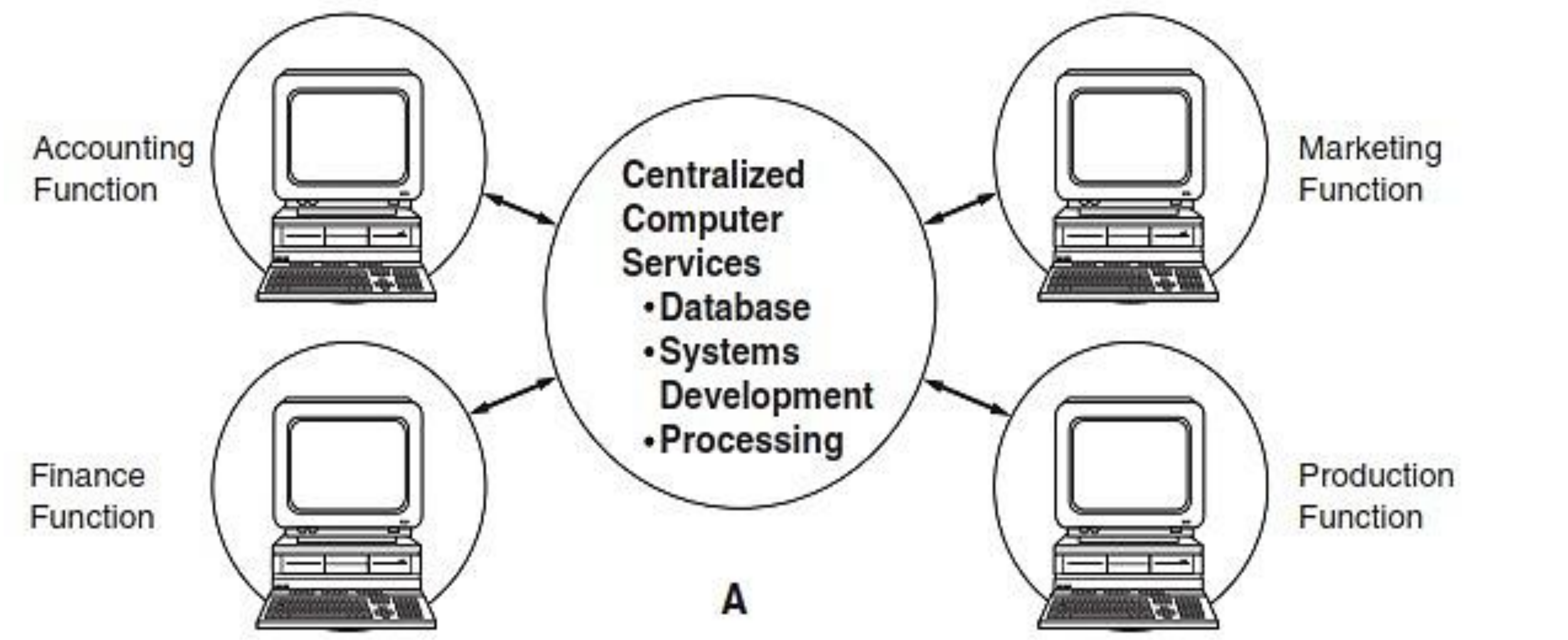
- All or any of the IT functions represented in centralized approach can be distributed
- The degree to which they are distributed will vary depending upon the philosophy and objectives of the organization's management
- There are two alternative DDP approaches
  - Approach A
  - Approach B

b) Distributed Data Processing

Approach A

- Alternative A is actually a variant of the centralized model; the difference is that terminals (or microcomputers) are distributed to end users for handling input and output
- This eliminates the need for the centralized data conversion groups, since the user now performs this tasks
- Under this model, however, systems development, computer operations, and database administration remain centralized

b) Distributed Data Processing

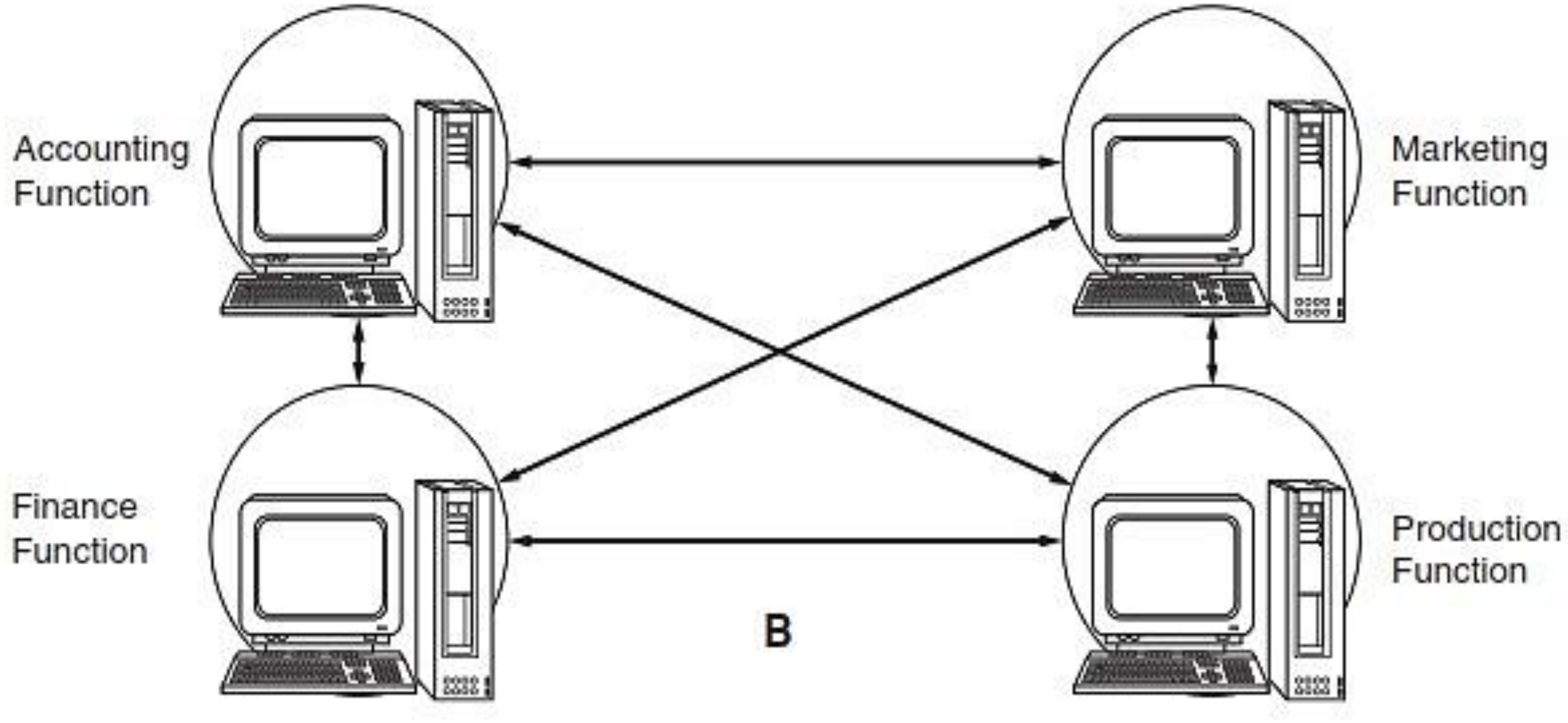


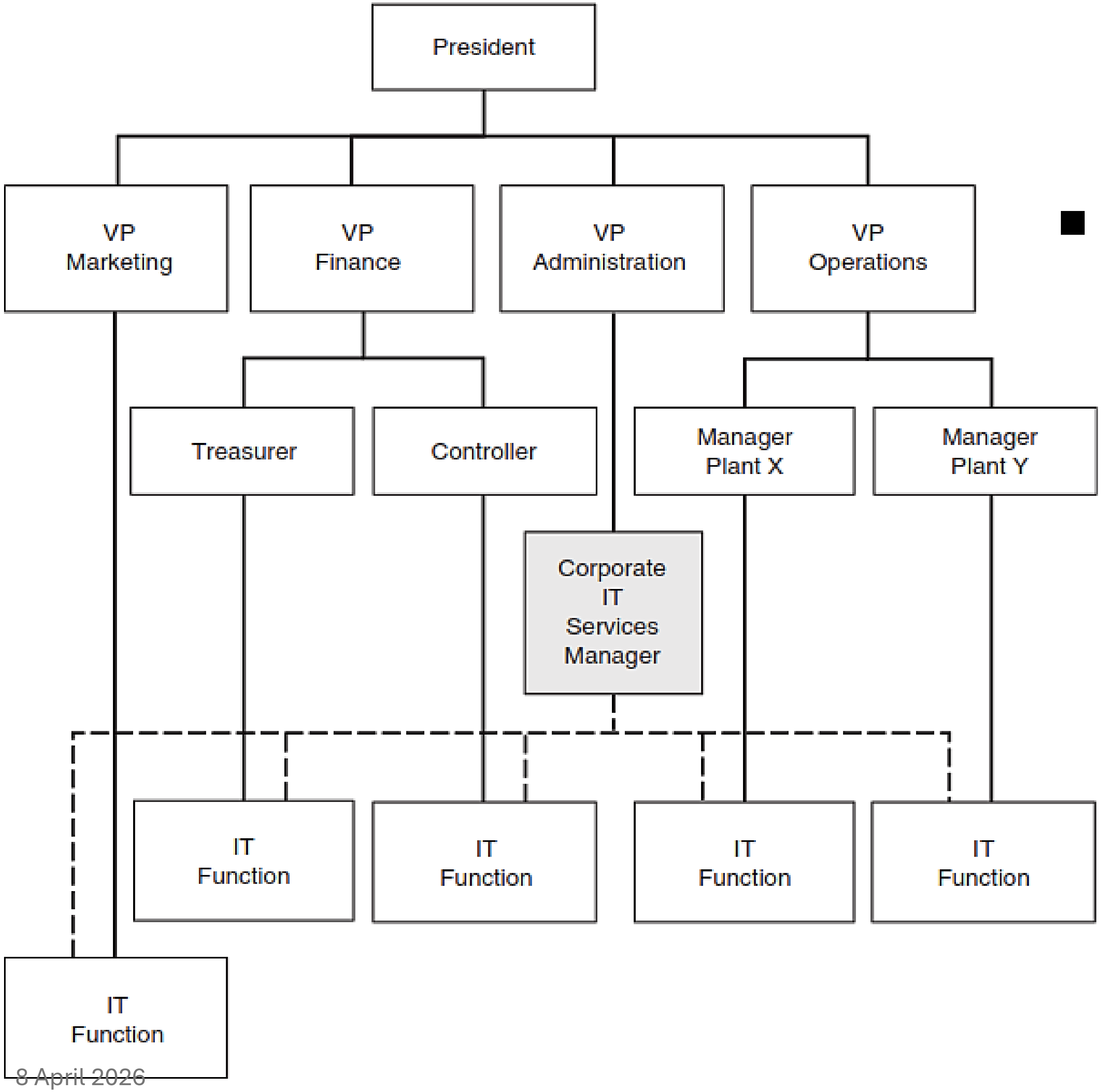
b) Distributed Data Processing

Approach B

- Alternative B is a significant departure from the centralized model
- This alternative distributes all computer services to the end users, where they operate as standalone units
- The result is the elimination of the central IT function from the organizational structure

b) Distributed Data Processing





- Possible organizational structure reflecting the distribution of all traditional data processing tasks to end-user areas

b) Distributed Data Processing

- Risks associated with DDP
  - Potential problems include
    i. **The inefficient use of resources,**
    ii. **The destruction of audit trails,**
    iii. **Inadequate segregation of duties,**
    iv. Increased potential for programming errors and systems failures,
    v. The lack of standards

b) Distributed Data Processing

i. The inefficient use of resources

- DDP can expose an organization to three types of risks associated with inefficient use of organizational resources

1. The risk of mismanagement of organization-wide IT resources by end users
2. DDP can increase the risk of operational inefficiencies because of redundant tasks being performed within the end-user committee
3. DDP environment poses a risk of incompatible hardware and software among end-user functions

b) Distributed Data Processing

ii. The destruction of audit trails

- An audit trail provides the linkage between a company's financial activities (transactions) and the financial statements that report on those activities
- Auditors use the audit trail to trace selected financial transactions from the source documents that captured the events, through the journals, subsidiary ledgers, and general ledger accounts that recorded the events, and ultimately to the financial statement themselves
- The audit trail is critical to the auditor's attest service

b) Distributed Data Processing

ii. The destruction of audit trails

- In DDP systems, the audit trail consists of a set of digital transaction files and master files that reside in part or entirely on end-user computers
- Should an end user inadvertently delete one of the files, the audit trail could be destroyed and unrecoverable
- Similarly, if an end user inadvertently inserts transaction errors into an audit trail file, it could become corrupted

b) Distributed Data Processing

iii. Inadequate segregation of duties

- Achieving an adequate segregation of duties may not be possible in some distributed environments
- For example, within a single unit the same person may write application programs, perform program maintenance, enter transaction data into the computer, and operate the computer equipment
- Such a situation would be a fundamental violation of internal control

b) Distributed Data Processing

- Advantages of DDP
    i. Cost reductions,
    ii. Improved cost control,
    iii. Improved user satisfaction,
    iv. Backup

Summary

- IT governance ensures IT supports business goals
- Organizational structure affects internal controls and auditing
- Centralized systems provide better control
- Distributed systems provide flexibility but increase risks
- Proper segregation of duties is essential for security and control

Thank you